Retrieval-Augmented Generation (RAG)

Large language models can write fluent text, but they only know what was in their training data. Retrieval-augmented generation fixes this by searching a document collection first and giving the retrieved passages to the model together with the question. The model then answers from those passages. This reduces made-up answers and lets the system cite its sources.

A RAG system has two stages. In the indexing stage, documents are split into chunks, each chunk is turned into an embedding vector, and the vectors are stored in a vector index. In the query stage, the question is embedded, the nearest chunks are retrieved, and they are placed into the prompt.

Improving retrieval quality

Hybrid search combines keyword matching with vector similarity. Keyword search is precise for names and rare terms, while vector search finds passages that express the same idea in different words. Reciprocal rank fusion merges both rankings without needing to compare their scores directly.

Agentic RAG lets an autonomous agent decide when to search, which query to use, and whether the evidence is good enough to answer. Such an agent may rewrite the query or search again after reading the first results.